Queen Mary
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EBU5376
Digital Signal Processing

Aliasing

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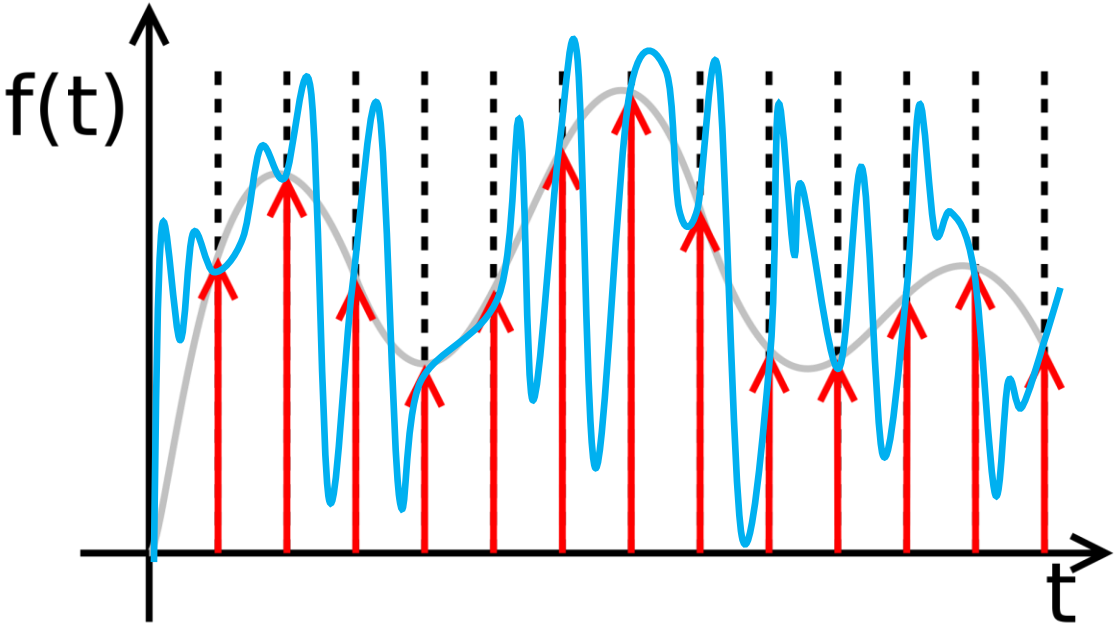
Questions to Answer

- Previous lecture discussed <u>how to obtain the frequency spectrum of analog signals</u> for subsequent signal processing
- Real world (analog) signal can't be stored/processed on computer
- In practical systems, signal processing is performed based on digital signals (sampled and quantized)

- *How do we ensure no loss of information in the conversion process?*
- *If a signal is sampled, what will happen to the frequency spectrum?*

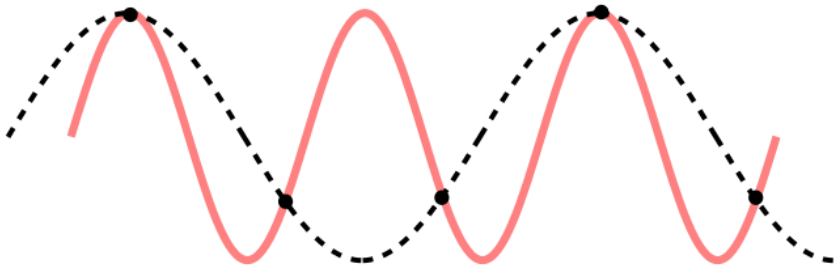
Problems with sampling



- Our red sample was good for the grey curve.
- What about the light blue curve?
- Blue same as the light grey at sample points.
- Problems if the original signal changes “fast”
- Can we be more precise what “fast” means?

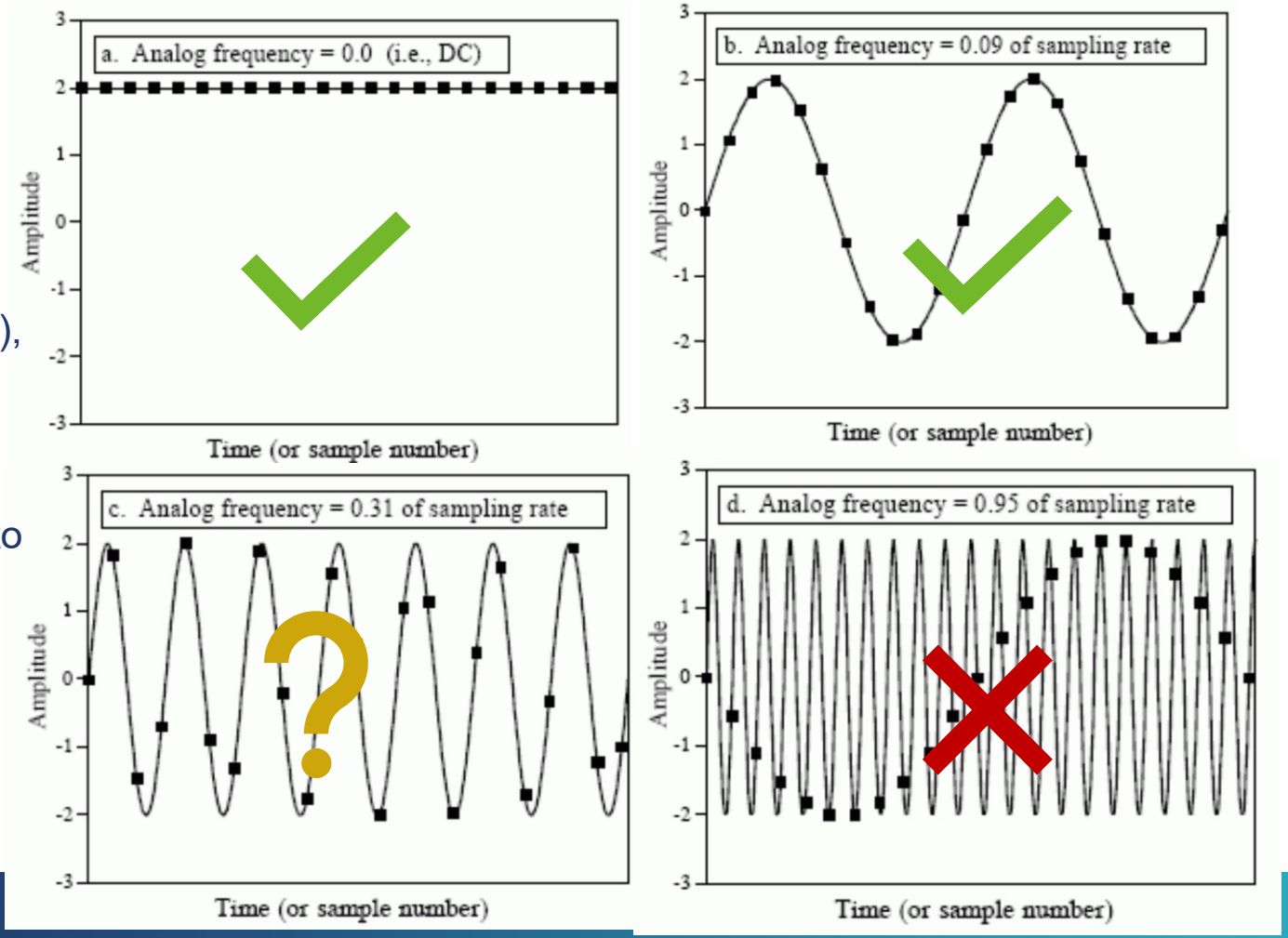
Outline

- Nyquist-Shannon Theorem
  - Nyquist frequency
- Aliasing
  - Finding aliasing frequency
- Bandlimiting



- Figure shows different signals under the same sampling rate
- As the signal oscillates faster (higher frequency), the sampled signal and the true signal become less alike
- Question: Is it possible to recover these signals based on the sample?

  Not if the signal frequency is too high.



Nyquist-Shannon Theorem

- Theorem: To ensure perfect reconstruction, a signal must be sampled at a rate greater than twice the highest frequency in the signal.

- If the highest frequency component is B Hz, then a sufficient sampling rate is $f_s > 2B$
- The threshold $f_s = 2B$ is the Nyquist rate
- In other words, for a fixed sampling rate f_s, this is enough for any signal with a maximum frequency component $B < \frac{f_s}{2}$.
- The frequency $B = \frac{f_s}{2}$ is the Nyquist frequency

*Theorem proof can be found in the Appendix

Example 1

- A signal is made up of frequencies 1000 Hz, 1800 Hz, and 2000 Hz. What is the Nyquist rate for sampling this signal?

Example 2

- If a signal is thought to have a frequency range between 1000 Hz and 4000 Hz, which of the following would be the most appropriate sample rate?

  a. 500 Hz

  b. 8000 Hz

  c. 9000 Hz

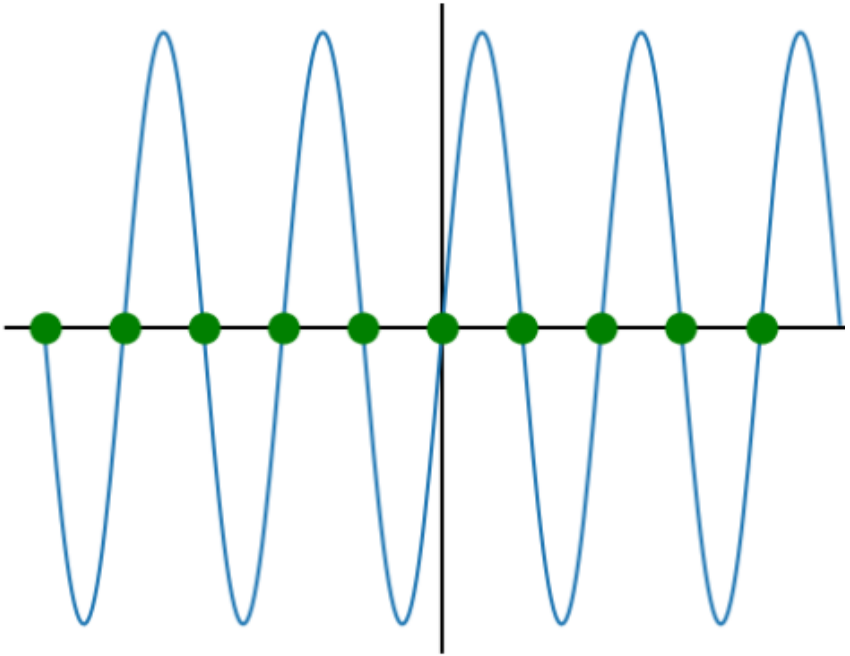
  d. 24000 Hz

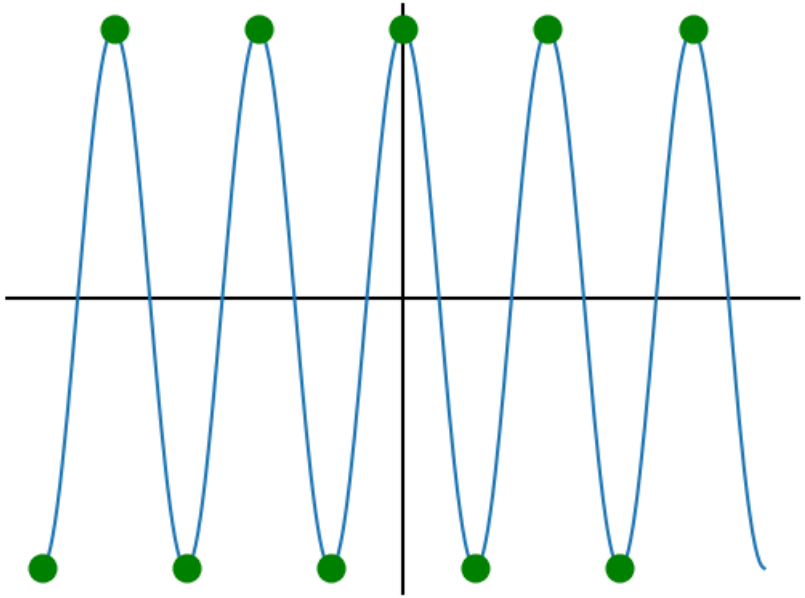
Example 3

- Signals can be modeled by sine waves, such as $x(t) = \sin(100\pi t)$. For this signal $x(t)$, is it appropriate to utilize a sample rate of 100 Hz? Why?

Example 3

Sampling at exactly 2B





Examples

1. A signal has frequencies up to 20Hz. How often must we sample?

2. We sample every 0.2 seconds. What is the Nyquist frequency? What is the maximum bandwidth we can sample?

3. The Nyquist frequency is 2Hz. How often are we sampling? Is this OK to sample a 3Hz signal?

Why Nyquist frequency is exactly two times the signal's maximum frequency?

What is happening to the frequency spectrum during sampling?

Frequency Spectrum of Periodic Signals

- In the previous lecture, we used the convolution theorem to show that convolving the signal $x(t)$ with an impulse train (periodization) corresponds to multiplying the spectrum $X(\omega)$ with an impulse train (sampling):

  Periodization in Time ↔ Sampling in Frequency

$$x(t) * \sum_{n=-\infty}^{\infty} \delta(t - nT_0) \leftrightarrow \frac{1}{T_0} \sum_{k=-\infty}^{\infty} X(\omega)\delta(\omega - k\omega_0)$$

  where $\omega_0 = \frac{2\pi}{T_0}$ is the fundamental angular frequency
- Convolution theorem: convolution in one domain ↔ multiplication in the other domain
- If instead we multiply in time, in the frequency domain we get convolution
- In other words, periodizing in one domain ↔ sampling in the other domain

Frequency Spectrum of Sampled Signals

- Multiplying the signal $x(t)$ with an impulse train yields a sampled signal

$$x_{sampled}(t) = x(t) \cdot \sum_{n=-\infty}^{\infty} \delta(t - nT_0) = \sum_{n=-\infty}^{\infty} x(t - nT_0)$$

Here T_0 is the sampling interval

- The Fourier transform of the impulse train is

$$\mathcal{F}\left\{\sum_{n=-\infty}^{\infty} \delta(t - nT_0)\right\} = \frac{2\pi}{T_0} \sum_{k=-\infty}^{\infty} \delta(\omega - k\omega_0)$$

- Let $X(\omega)$ denotes the spectrum of the signal $x(t)$, the Fourier transform of $x_{sampled}(t)$ is

$$\mathcal{F}\left\{x(t) \cdot \sum_{n=-\infty}^{\infty} \delta(t - nT_0)\right\} = \mathcal{F}\{x(t)\} * \mathcal{F}\left\{\sum_{n=-\infty}^{\infty} \delta(t - nT_0)\right\} = X(\omega) * \frac{2\pi}{T_0} \sum_{k=-\infty}^{\infty} \delta(\omega - k\omega_0)$$

- This convolution effectively shifts the spectrum $X(\omega)$ to multiples of $\omega_0 = \frac{2\pi}{T_0}$
- Thus, the spectrum $X(\omega)$ is repeated periodically at intervals of $\omega_0 = \frac{2\pi}{T_0}$

Frequency Spectrum of Sampled Signals

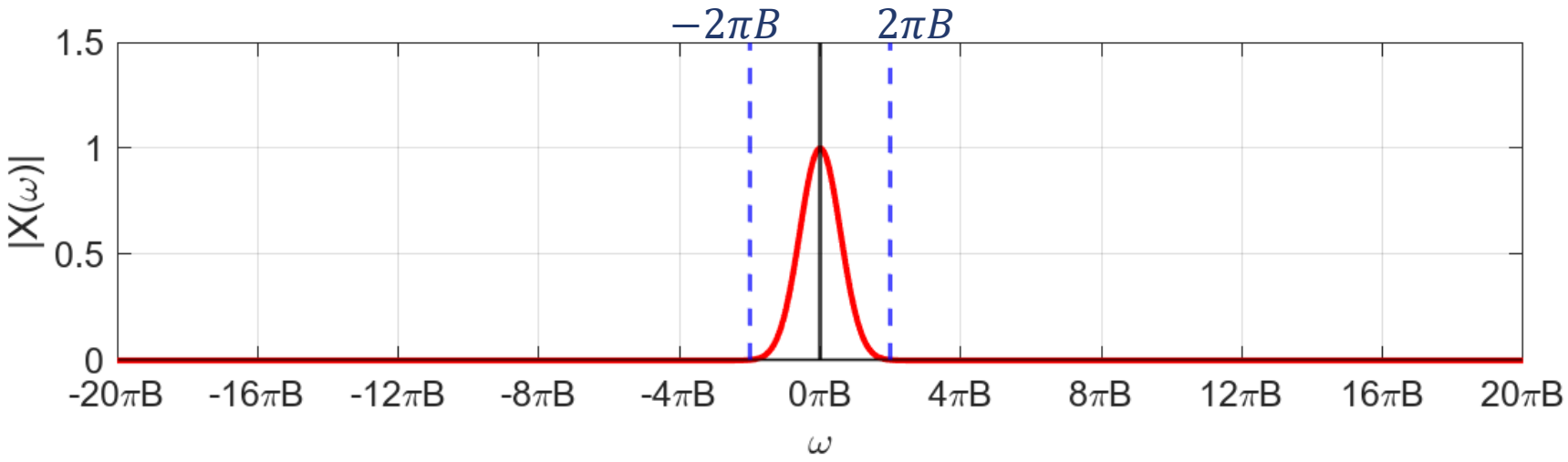
Sampling in Time ↔ Periodization in Frequency

- When you **sample** a signal every T_0 seconds, the resulting spectrum becomes **periodic** in frequency with a fundamental period of $\omega_0 = \frac{2\pi}{T_0}$

- Increase T_0 (i.e., **lower sampling rate in time**)
  - ω_0 decreases
  - The frequency-domain copies (repetitions) appear more frequently (**closer together**).
- Decrease T_0 (i.e., **higher sampling rate in time**)
  - ω_0 increases
  - The frequency-domain copies are **more widely spaced**

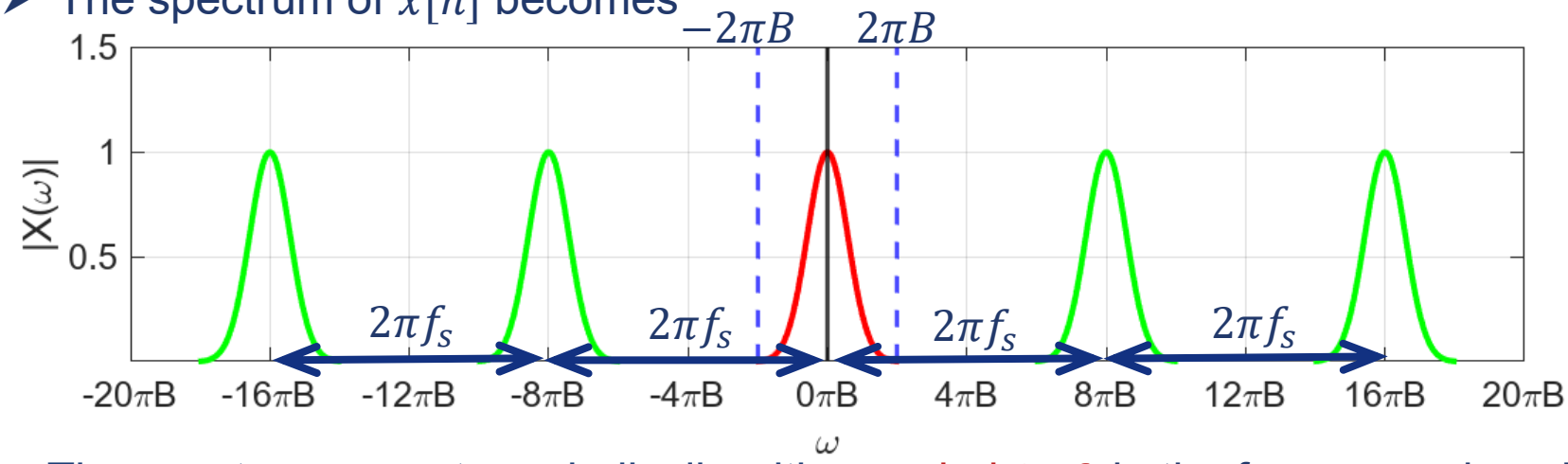
Frequency Spectrum of Sampled Signals

- Suppose $x(t)$ is a continuous signal and B Hz is the maximum frequency component in $x(t)$.
- The spectrum of $x(t)$ can be visualized as



Frequency Spectrum of Sampled Signals

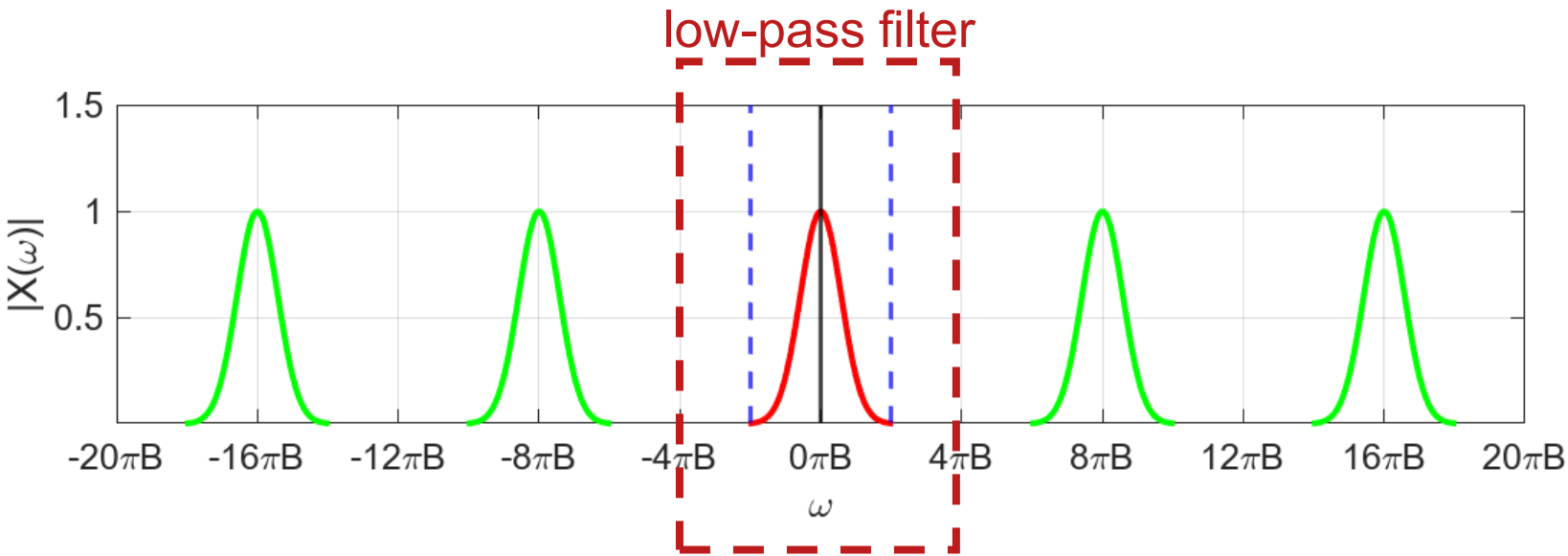
- Now, we sample $x(t)$ at $f_s = \frac{1}{T_0} = 4B$ Hz. The sampled signal is denoted as $x[n]$.
- The spectrum of $x[n]$ becomes



- The spectrum repeats periodically with a period $2\pi f_s$ in the frequency domain
- The original spectrum of the signal (in red) is known as the baseband, representing the frequency range where the essential information of the signal is contained

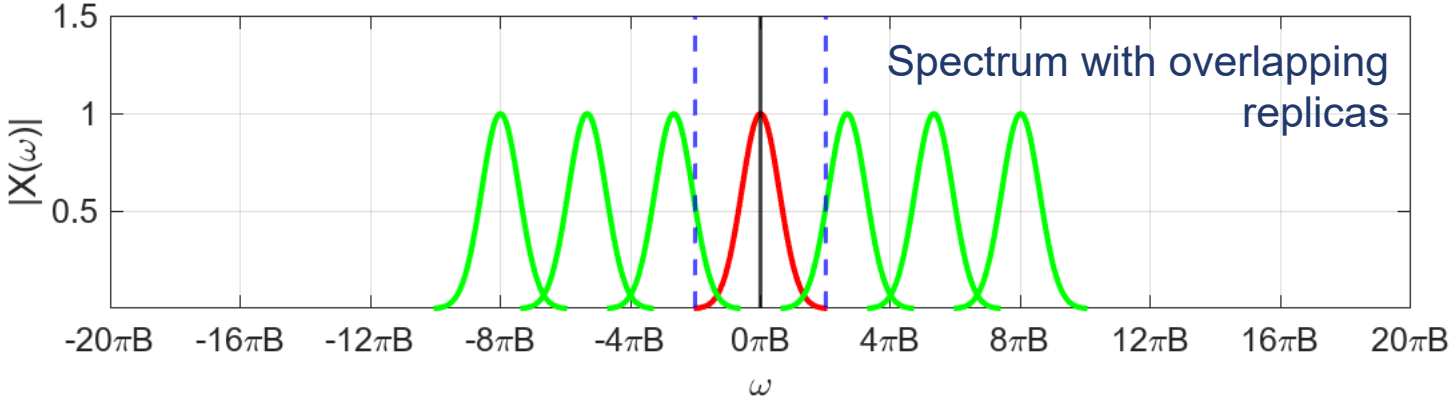
Frequency Spectrum of Sampled Signals

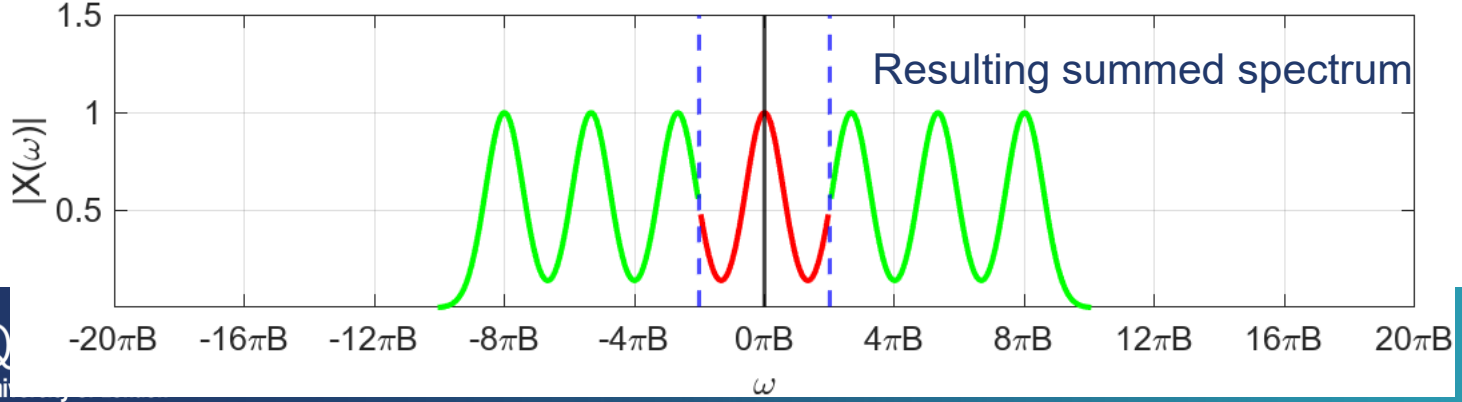
- To reconstruct the original signal $x(t)$, an ideal low-pass filter can be applied to isolate the baseband spectrum and remove the high-frequency replicas



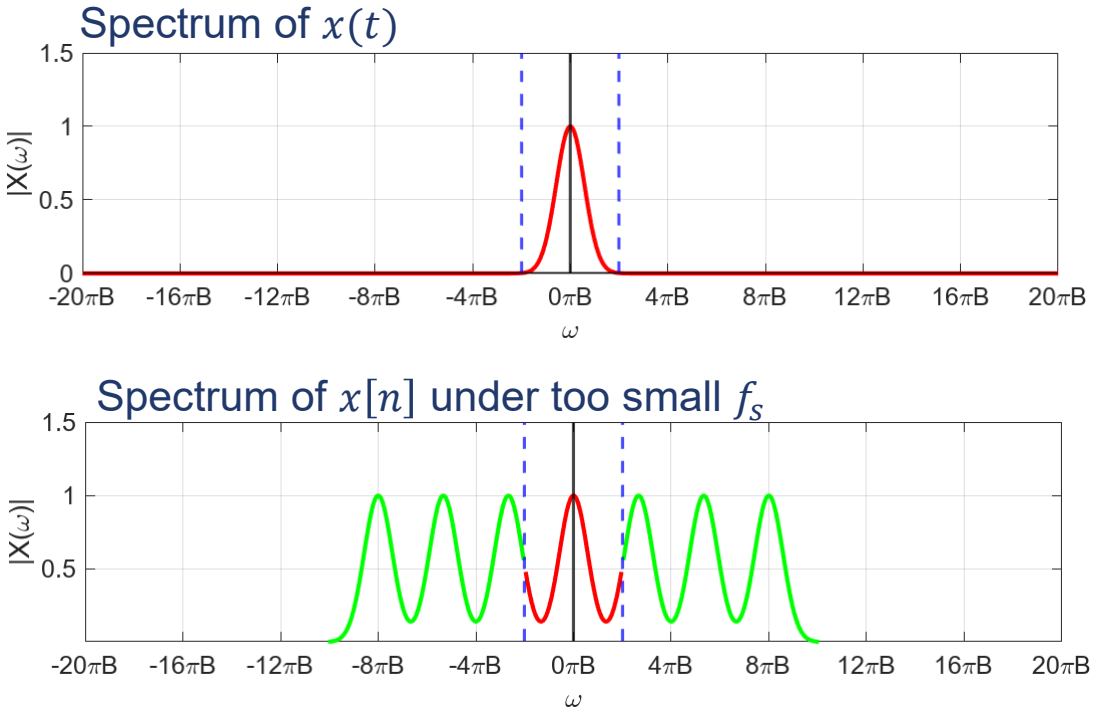
Frequency Spectrum of Sampled Signals

- If the sampling rate is too low, i.e., f_s very small, the replicas overlap with the baseband



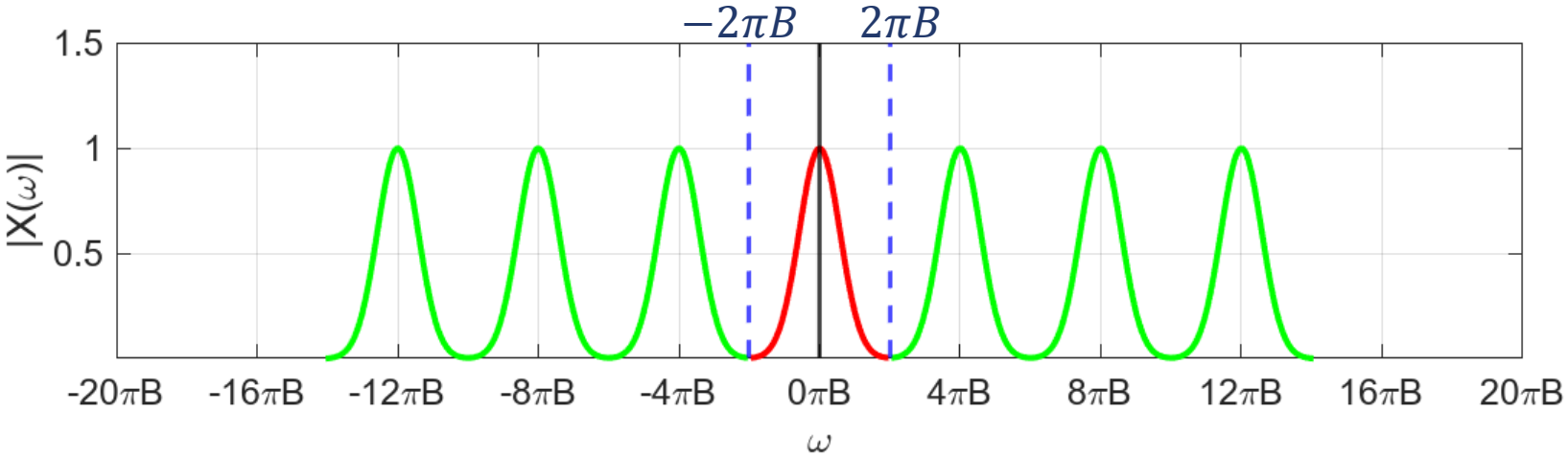


Frequency Spectrum of Sampled Signals



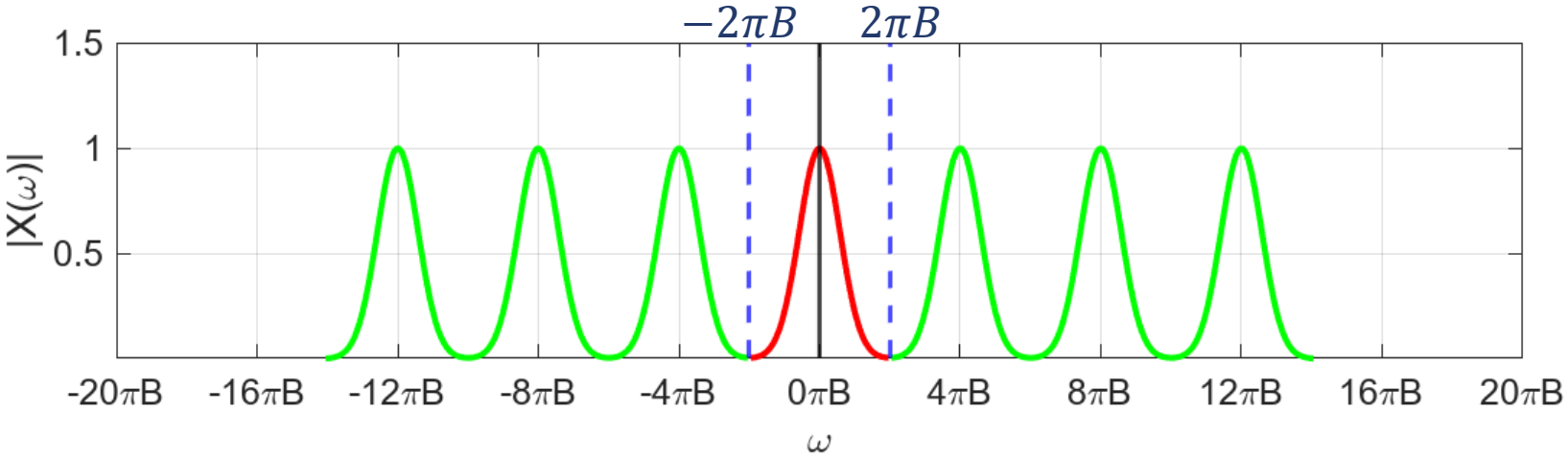
- This overlap results in high-frequency components (e.g., beyond $2\pi B$) being *folded back* into the baseband range, altering the spectrum within $-2\pi B$ to $2\pi B$
- Perfect reconstruction of the original continuous-time signal becomes impossible
- This results in signal aliasing

Frequency Spectrum of Sampled Signals



- Replicas begin to overlap when $f_s = 2B$
- To avoid overlapping (aliasing), we need $f_s > 2B$, i.e., the sampling rate must be greater than twice the signal bandwidth
- This threshold $f_s = 2B$ is the Nyquist rate

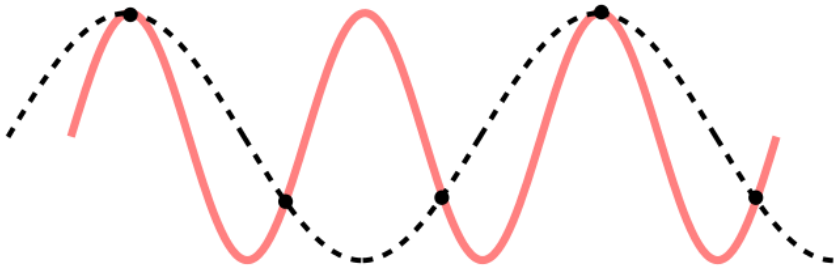
Frequency Spectrum of Sampled Signals



- Similarly, for a given sampling rate f_s, the signal bandwidth B defines the width of the baseband and the replicas
- A bandwidth B that is too large causes overlapping spectra (aliasing). To avoid this, we need $B < \frac{f_s}{2}$, i.e. the signal bandwidth cannot exceed half of the sampling rate
- This threshold $B = \frac{f_s}{2}$ is the Nyquist frequency

Outline

- Nyquist-Shannon Theorem
  - Nyquist frequency
- Aliasing
  - Finding aliasing frequency
- Bandlimiting

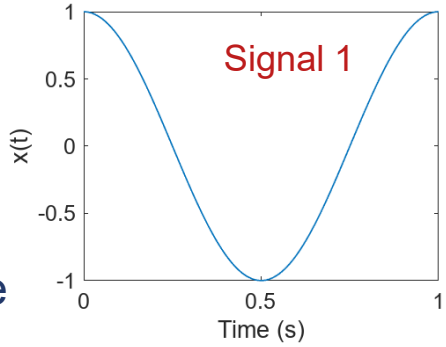


Aliasing

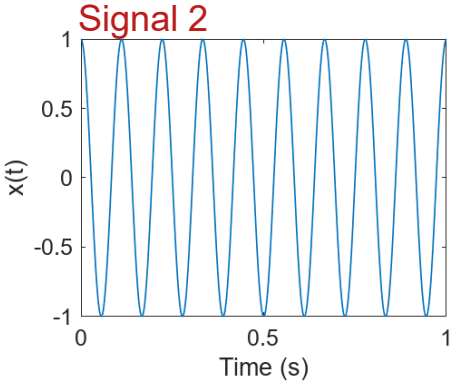
- What will happen in the time-domain if the signal frequency is higher than the Nyquist frequency?
- Let's try an experiment

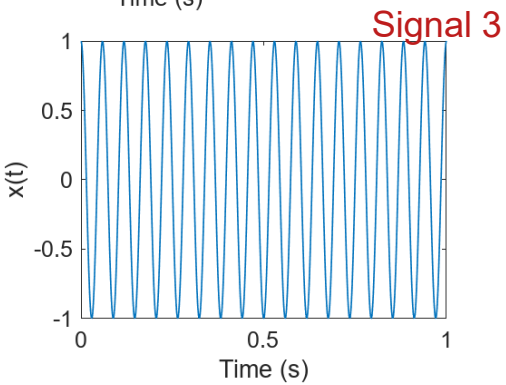
  Sampling rate is fixed to 8Hz in this experiment

  We will investigate the samples obtained from signals of different frequencies under this sampling rate



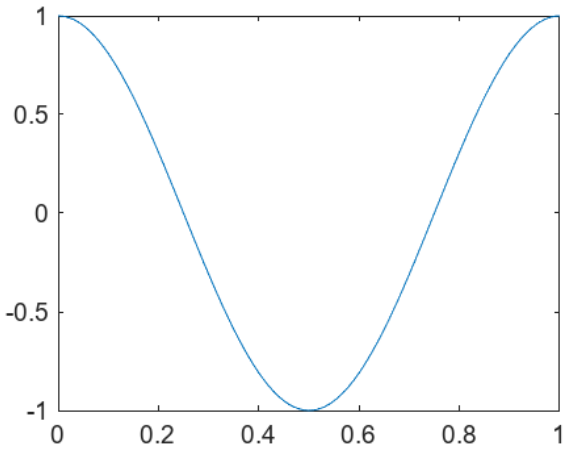
- **Experiment (3 signals)**
- Signal 1: $x_1(t) = \cos(2\pi t)$
- Signal 2: $x_2(t) = \cos(18\pi t)$
- Signal 3: $x_3(t) = \cos(34\pi t)$



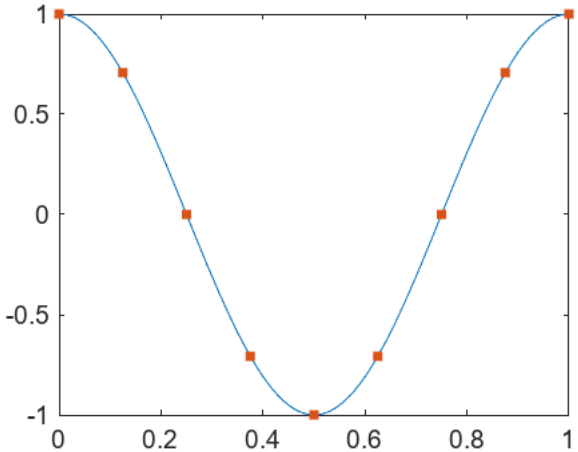


Aliasing – Experiment

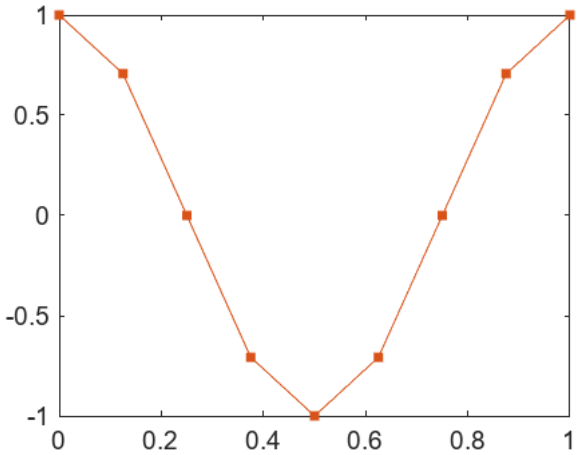
- Sampling rate is 8Hz
- Signal 1: $x_1(t) = \cos(2\pi t)$



Analog waveform



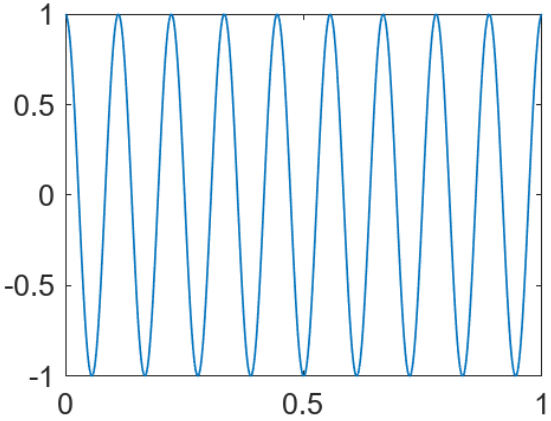
Sampling



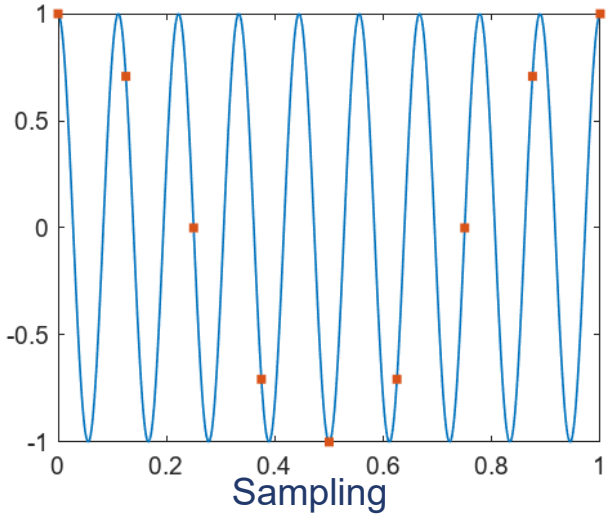
Sample-based Waveform

Aliasing – Experiment

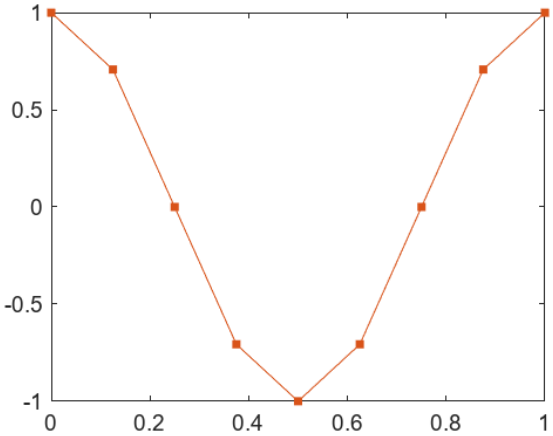
- Sampling rate is 8Hz
- Signal 2: $x_2(t) = \cos(18\pi t)$



Analog waveform



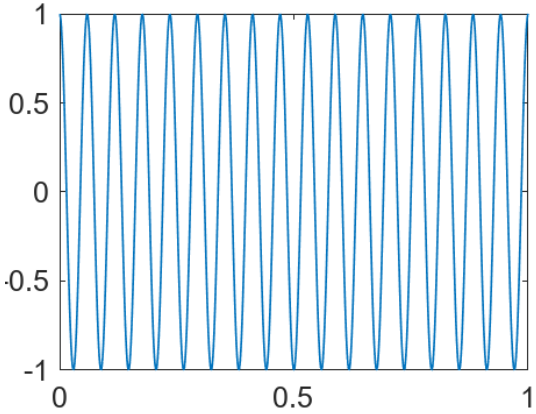
Sampling



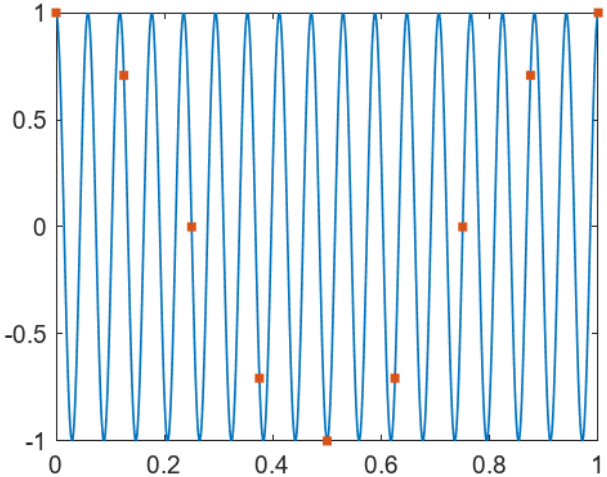
Sample-based Waveform

Aliasing – Experiment

- Sampling rate is 8Hz
- Signal 3: $x_3(t) = \cos(34\pi t)$

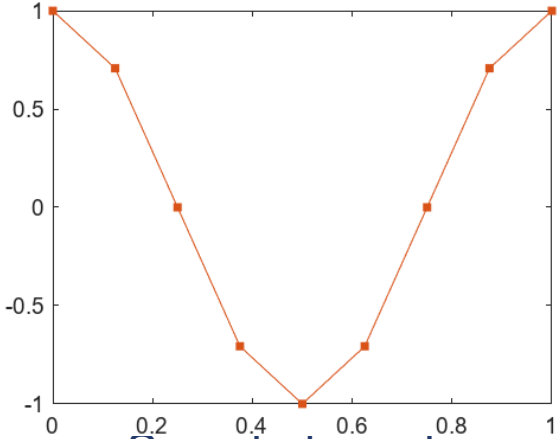


Analog waveform



Sampling

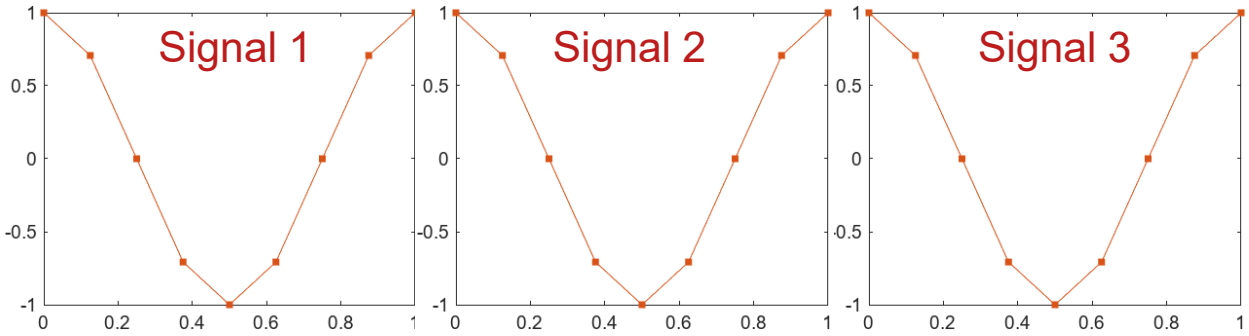
Seeing a pattern?



Sample-based Waveform

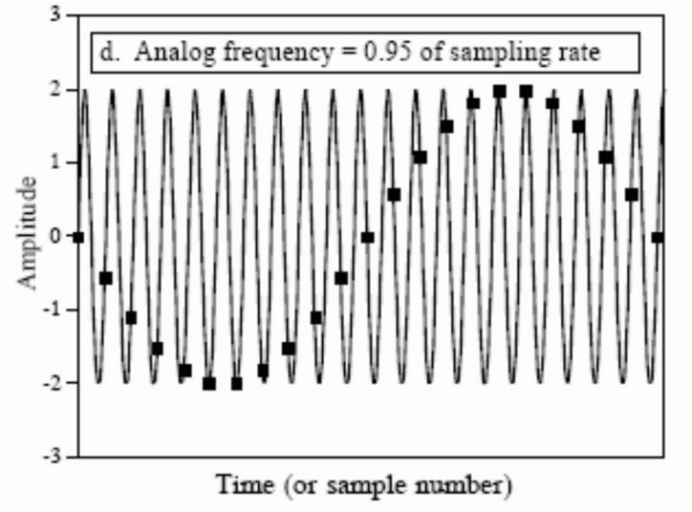
Aliasing – Experiment

- Sampling rate is 8Hz
- Signal 1: $x_1(t) = \cos(2\pi t)$
- Signal 2: $x_2(t) = \cos(18\pi t)$
- Signal 3: $x_3(t) = \cos(34\pi t)$



- **Observation:**
- The SAME set of samples are obtained from three signals at 8 Hz sampling rate
- If only the samples are given, they all appear identical to the samples of Signal 1
- This is because 8Hz is a too low sampling rate for Signal 2 and 3, according to Nyquist
- This demonstrates a typical case of aliasing

Aliasing

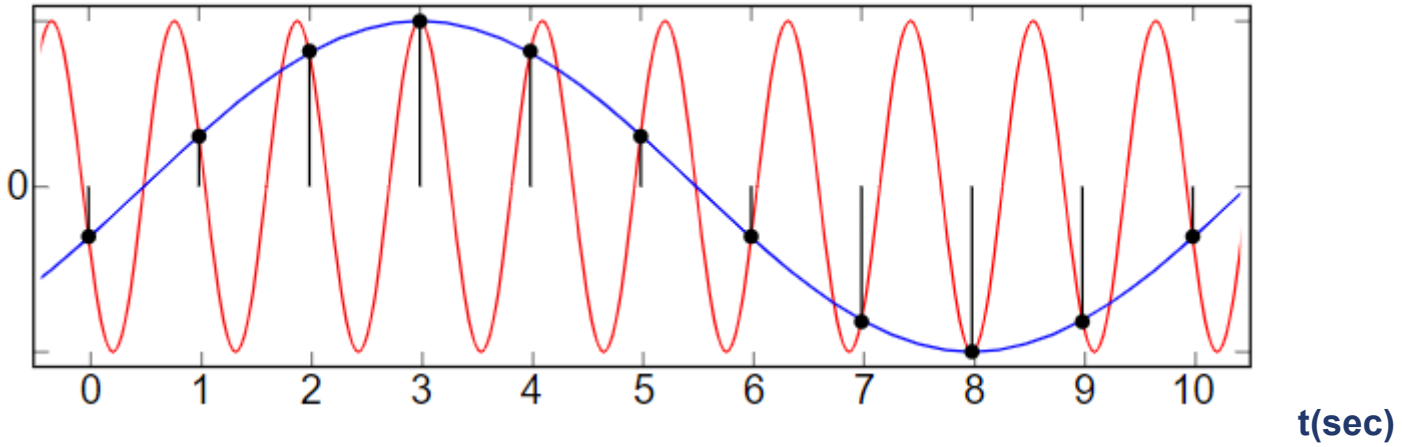


- **Aliasing** is the effect of new frequencies appearing in the sampled signal, that were not present in the original signal.

- It is caused by **undersampling, i.e.,**
  - a too low sampling rate that is <u>less than Nyquist rate</u> of the given signal

  or
  - a too high signal frequency that <u>exceeds Nyquist frequency</u> of the sampling system

Aliasing



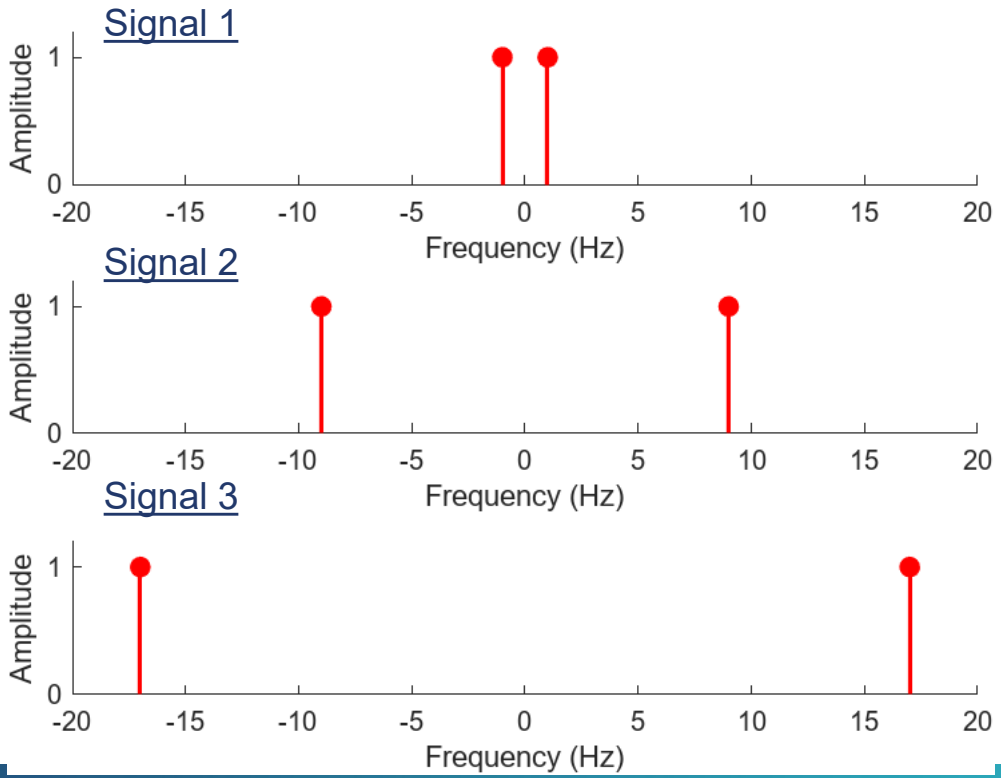
- Red line is sine wave at 0.9Hz
- Blue line is a sine wave at 0.1Hz
- Black dots are samples separated by T_s=1sec.
- We say 0.1Hz and 0.9Hz are aliases under sampling rate 1Hz
- In general, *how to find the aliases*?

Aliasing

In the previous experiment, we discovered that the following three signals have the same set of samples under a sampling rate of 8 Hz

- Signal 1: $x_1(t) = \cos(2\pi t)$
- Signal 2: $x_2(t) = \cos(18\pi t)$
- Signal 3: $x_3(t) = \cos(34\pi t)$

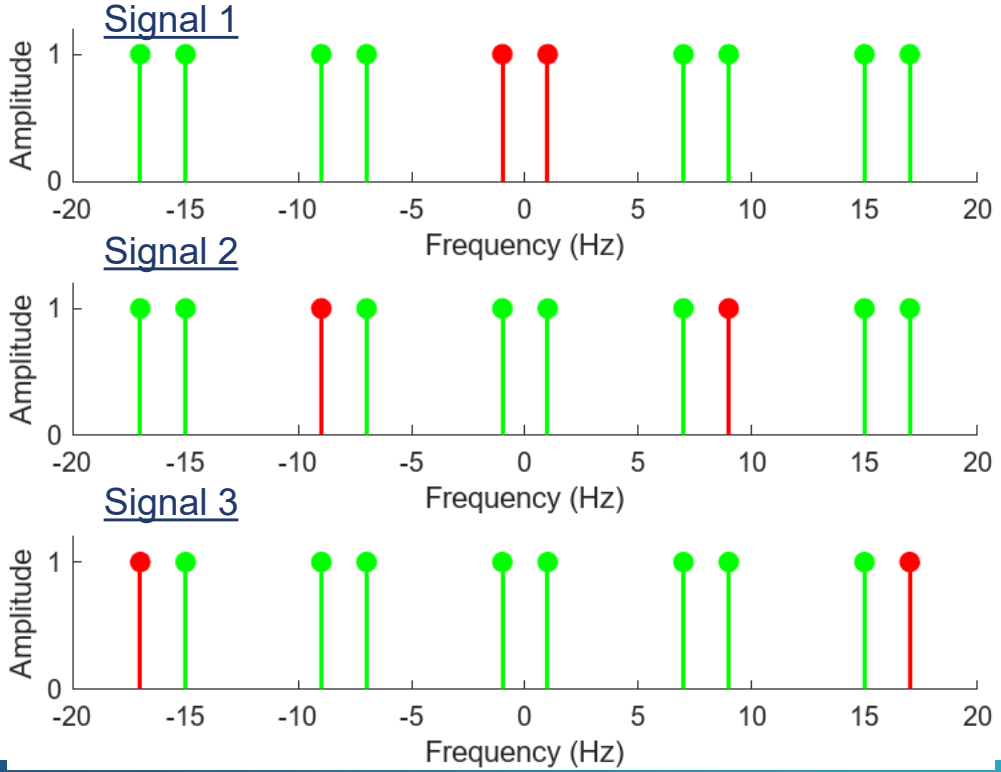
Before sampling, their spectrums are:



Aliasing

- Signal 1: $x_1(t) = \cos(2\pi t)$
- Signal 2: $x_2(t) = \cos(18\pi t)$
- Signal 3: $x_3(t) = \cos(34\pi t)$

After sampling at 8Hz, their spectrums become **identical**:



In fact, all frequencies with non-zero amplitudes on this spectrum are aliased to each other under 8Hz sampling rate. These includes $\pm 1, \pm 7, \pm 9, \pm 15, \pm 17$Hz,…

There is a general equation to derive these aliasing frequencies

Aliasing

- Given an under-sampled real-valued signal, it is aliased to the following frequencies

$$f_a = |nf_s - f| \ , \qquad n = 0, \pm 1, \pm 2, \ldots$$

- In the equation, f_s is the sampling rate, f is the (fundamental) signal frequency
- Usually, we are only interested in the lowest positive frequency that the signal is aliased to
  - The lowest frequency is the one that can be sampled without aliasing

Aliasing Examples

Example 1: Given a system with sampling rate 44 Hz and a signal of frequency 32 Hz, determine whether the signal is under-sampled. If so, find the lowest frequency that it is aliased to.

Aliasing Examples

<u>Example 2</u>: Given the sinusoid $x(t) = \sin(2\pi t)$ and an 8Hz sampling rate, find the lowest 5 frequencies that are aliased to $x(t)$

Aliasing Examples

Example 2: Given the sinusoid $x(t) = \sin(2\pi t)$ and an 8Hz sampling rate, find the lowest 5 frequencies that are aliased to $x(t)$

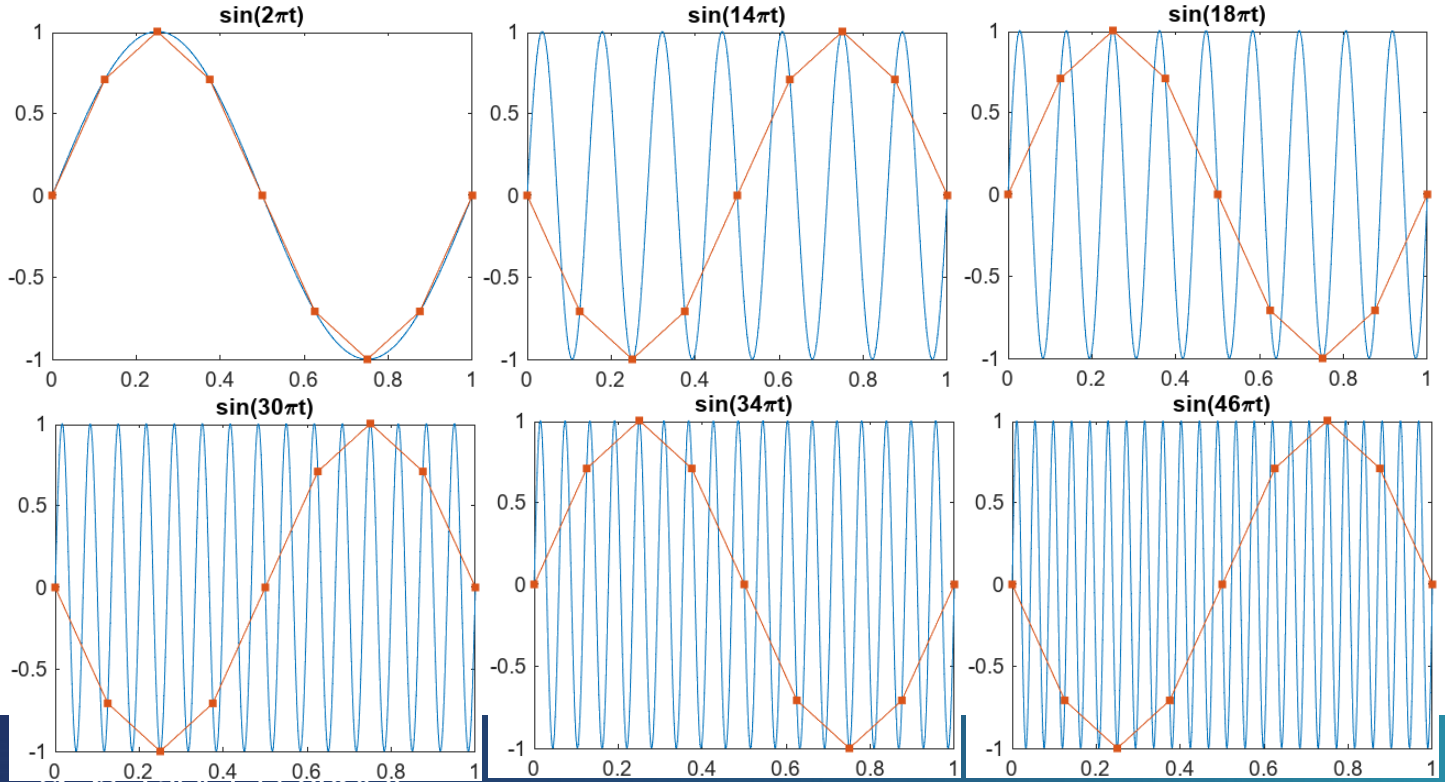
From last slide, we obtained $7Hz, 9Hz, 15Hz, 17Hz, 23Hz$ as the lowest 5 frequencies that are aliased to $x(t) = \sin(2\pi t)$

Does it mean we can write down the equations of these 5 signals?
E.g., The aliased signals are $\sin(14\pi t)$, $\sin(18\pi t)$, $\sin(30\pi t)$, $\sin(34\pi t)$, $\sin(46\pi t)$.

Let's check their samples

Aliasing Examples

Example 2: Figures show the samples of the 5 alias frequencies of $\sin(2\pi t)$ under 8Hz



The sample-based waveforms have:

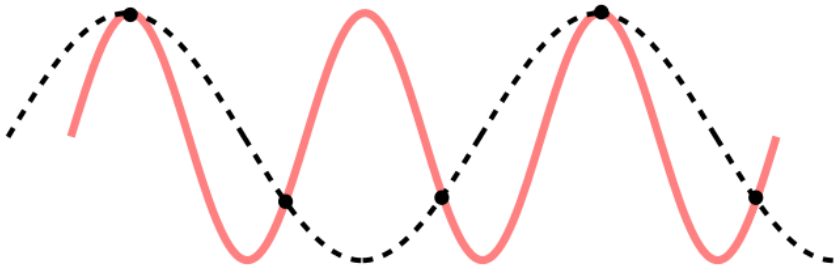
- ❑ *Same frequency*
- ❑ *Different phase*

Related (but not exactly aliasing)

- The helicopter blades rotate very quickly. Call its rotation period T_h.
- The video camera captures a still image every T_c.
- When $T_h = nT_c$ (or very nearly equal) the helicopter blades appear to be still or moving really slowly.
- This is, again, a sampling issue.

Outline

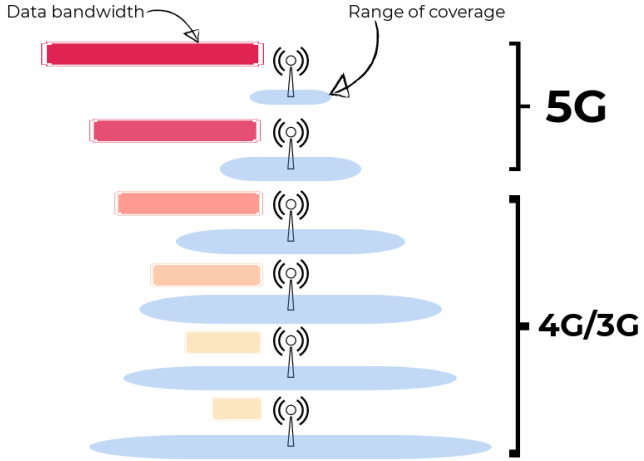
- Nyquist-Shannon Theorem
  - Nyquist frequency
- Aliasing
  - Finding aliasing frequency
- Bandlimiting



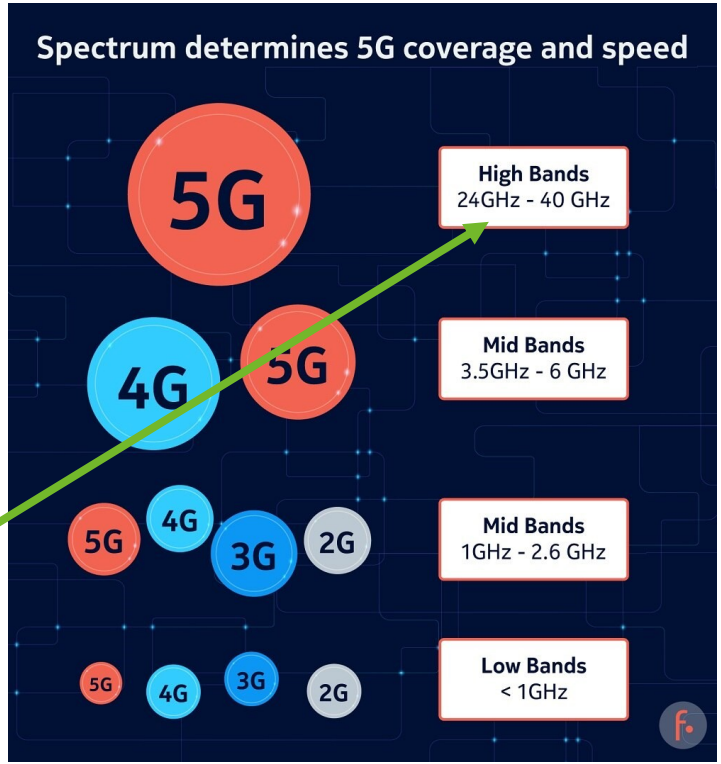
Bandlimiting

Definitions

- In telecommunication, a **band** means a range of frequencies in the radio electromagnetic spectrum
- A **bandwidth** is the difference between the upper and lower frequencies in a band



Bandwidth is 16 GHz



5G spectrum bands explained— low, mid and high band | Nokia

Bandlimiting

- Story so far:
  - If a signal has $B < f_s/2$, we can sample it with no information loss.
  - Otherwise, we get “aliasing”.
- What about real signals?
  - Typically they have no particular limit on their bandwidth.
  - Any information signal contains a large number of frequencies, so, to decide a sampling frequency is always a problem.
  - Even if we sample very fast some portion of the signal is still aliased.
- How can we solve this?
  - One option “bandlimiting”.
  - Filter our analog signal before sampling.

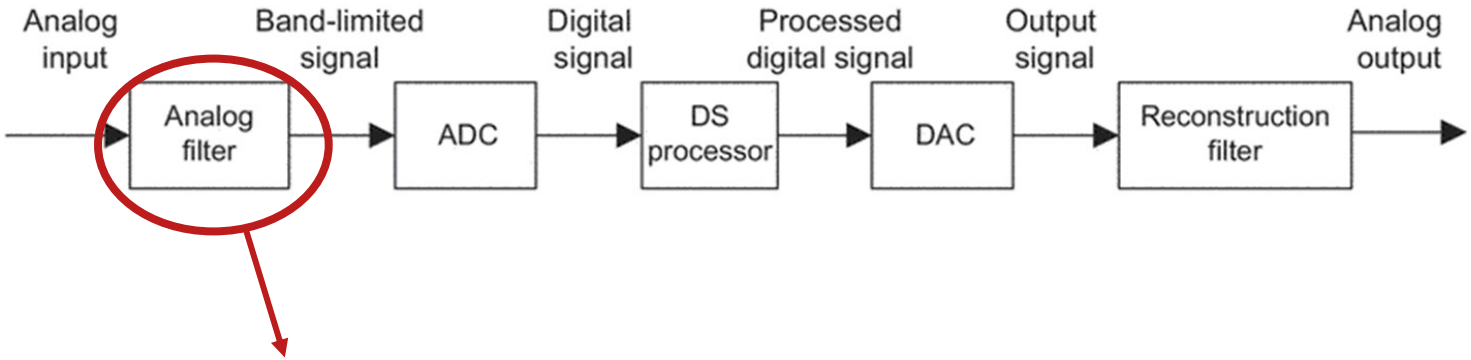
What is bandlimiting?

- We say a signal is "bandlimited" if it contains no frequencies above some frequency B, known as the bandwidth.
- If we can ensure a signal is "bandlimited" we can find some f_s which can sample it.
- An obvious strategy is to filter the signal after receiving it.

- If we pick an f_s then we must set a bandlimit of $\frac{f_s}{2}$ to avoid aliasing
- E.g., A low pass filter limiting frequencies above $\frac{f_s}{2}$ will remove aliasing into frequencies < $\frac{f_s}{2}$.

Bandlimiting

- Think back to this slide from lecture 1



- The “analog filter” is the band limiting.

Lecture summary

- **Nyquist-Shannon theorem**:
  - To sample a signal with bandlimit B, the sampling frequency f_s must satisfy $f_s > 2B$.
  - Half the sampling frequency is known as the Nyquist frequency $f_n = f_s/2$.
- **Aliasing**:
  - Signals will "alias" to signals with other frequencies.
  - Computed by $f_a = |n \cdot f_s - f|$
- **Bandlimiting**:
  - Filter out frequencies above f_n before sampling.
  - To avoid aliasing

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Appendix - Shannon proof

- We have a signal $x(t)$ with spectrum $X(\omega)$ which is zero except when $\left|\frac{\omega}{2\pi}\right| < B$ (that is within our bandwidth B).
- $x(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(\omega)e^{j\omega t}d\omega = \frac{1}{2\pi}\int_{-2\pi B}^{2\pi B} X(\omega)e^{j\omega t}d\omega$
- Let $t = n/2B$ for some integer n (positive or negative) then
- $x(n/2B) = \frac{1}{2\pi}\int_{-2\pi B}^{2\pi B} X(\omega)e^{j\omega n/2B}d\omega$
- Let $y[n] = x(n/2B)$ for all integer n then this is the same as saying y are samples taken once every 1/2B seconds
- $y[n] = \frac{1}{2\pi}\int_{-2\pi B}^{2\pi B} X(\omega)e^{j\omega n/2B}d\omega$ for all integers n

Appendix - Shannon proof

$$y[n] = \frac{1}{2\pi}\int_{-2\pi B}^{2\pi B} X(\omega)e^{j\omega n/2B}d\omega$$

- Recall inverse DTFT -- use $Y(\omega)$ to distinguish from $X(\omega)$

$$y[n] = \frac{1}{2\pi}\int_{-\pi}^{\pi} Y(\omega)e^{j\omega n}d\omega$$

- If we substitute $\omega = \frac{\omega'}{2B}$ this is

$$y[n] = \frac{2B}{2\pi}\int_{-2B\pi}^{2B\pi} Y\left(\frac{\omega'}{2B}\right)e^{\frac{j\omega' n}{2B}}d\omega'$$

We can see then that $2BY\left(\frac{\omega}{2B}\right) = X(\omega)$

Appendix - Shannon proof

Since $2BY\left(\frac{\omega}{2B}\right) = X(\omega)$ we can say that the frequency spectrum $X(\omega)$ we want is just a rescaling of $\mathrm{Y}(\omega)$ in the frequency range $-2\pi B < \omega < 2\pi B$

However we assumed that $X(\omega)$ is 0 outside this range.

Hence we can completely reconstruct $X(\omega)$ from our samples y[n].

If we can completely reconstruct $X(\omega)$ we can completely reconstruct $x(t)$.